Communications System

Software Design Specification

Revision History

Date	Revision	Description	Author
03/22	1.0	Setting up the document	Shea Pettit
04/05	1.1	Added subsections, worked on purpose, and added diagrams	Everyone
04/07	1.2	Organized the diagrams, explaining the contents	Everyone
04/08	1.3	Finalizing the document	Everyone

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1. Purpose

This document outlines the architecture of the Communications System.

1.1. Scope

The Design Specification Document will provide an overview of the design decisions behind the Architecture of the Communication System. This document is intended for the developers of the system.

1.2. Definitions, Acronyms, Abbreviations

1.2.1. User Login Module - ULM
1.2.2. Client Module - CM
1.2.3. IT module - IM
1.2.4. Communication Server Module - CSM

1.3. References

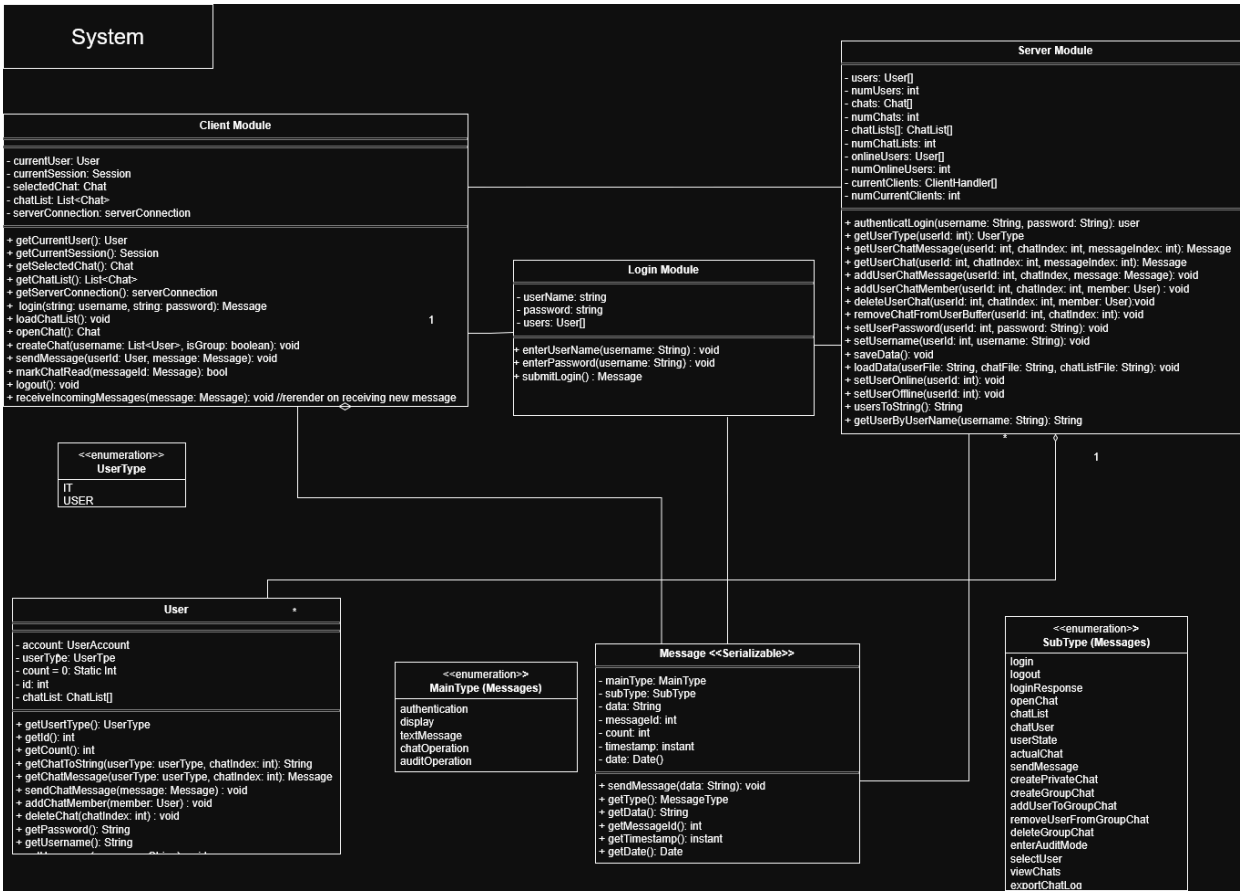
1.3.1. Software Requirements Specification PDF
1.3.2. Use Case Specification PDF

1.4. System Overview

Scalable, text only communications system designed to sustain sending messages between multiple users. These messages can be sent synchronously and asynchronously in group chats or private chats. Users will be limited to access to messages that only correspond to them, and all messages will be viewable by IT staff. Users and IT staff interact with the system through the provided GUI.

2. Architecture Design

2.1. Architecture Overview

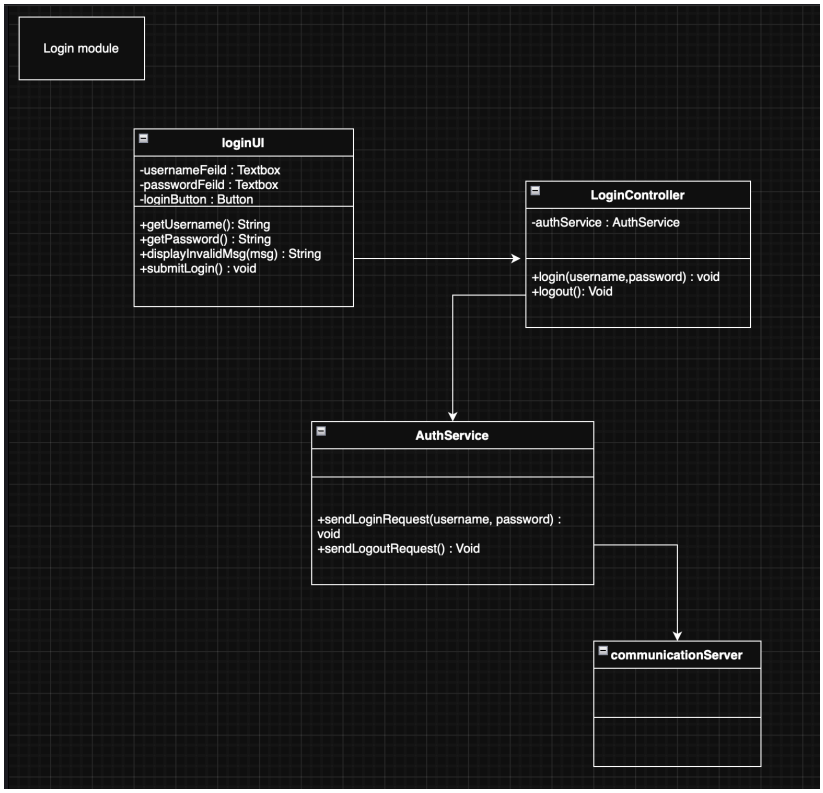


The communications system follows client-server architecture composed of four modules: The User Login Module(ULM), Client Module(CM), IT Module(IM), and Communication Server Module (CSM), along with User and Message entities. The login module is responsible for handling user credential authentication and sends requests that are in charge of validating the input. The Server module is the core component of the system and is responsible for managing authentication, users, chats, messages, and data handling. Once the server establishes and provides an authentication response, the user is granted access to interact with the Client Module. The Client module manages another core component of the system, the messaging and chat functionality. In addition to that, the client module handles communication to the server. The User encapsulates the account information and user type used to define access permissions. Additionally, it stores unique, identifiable information for each user along with the chats that they have access to. The message class represents the different types of communication, ranging from user to user, to client to server and server to client. The type of communication is labeled and distinguished by the MessageType assigned to each message created. This architecture supports synchronous and asynchronous messaging, private and

group chat creation, and IT audit access to chats. The system is designed to support authentication, chat management, messaging, and role-based permissions, entirely dependent on client-server interactions.

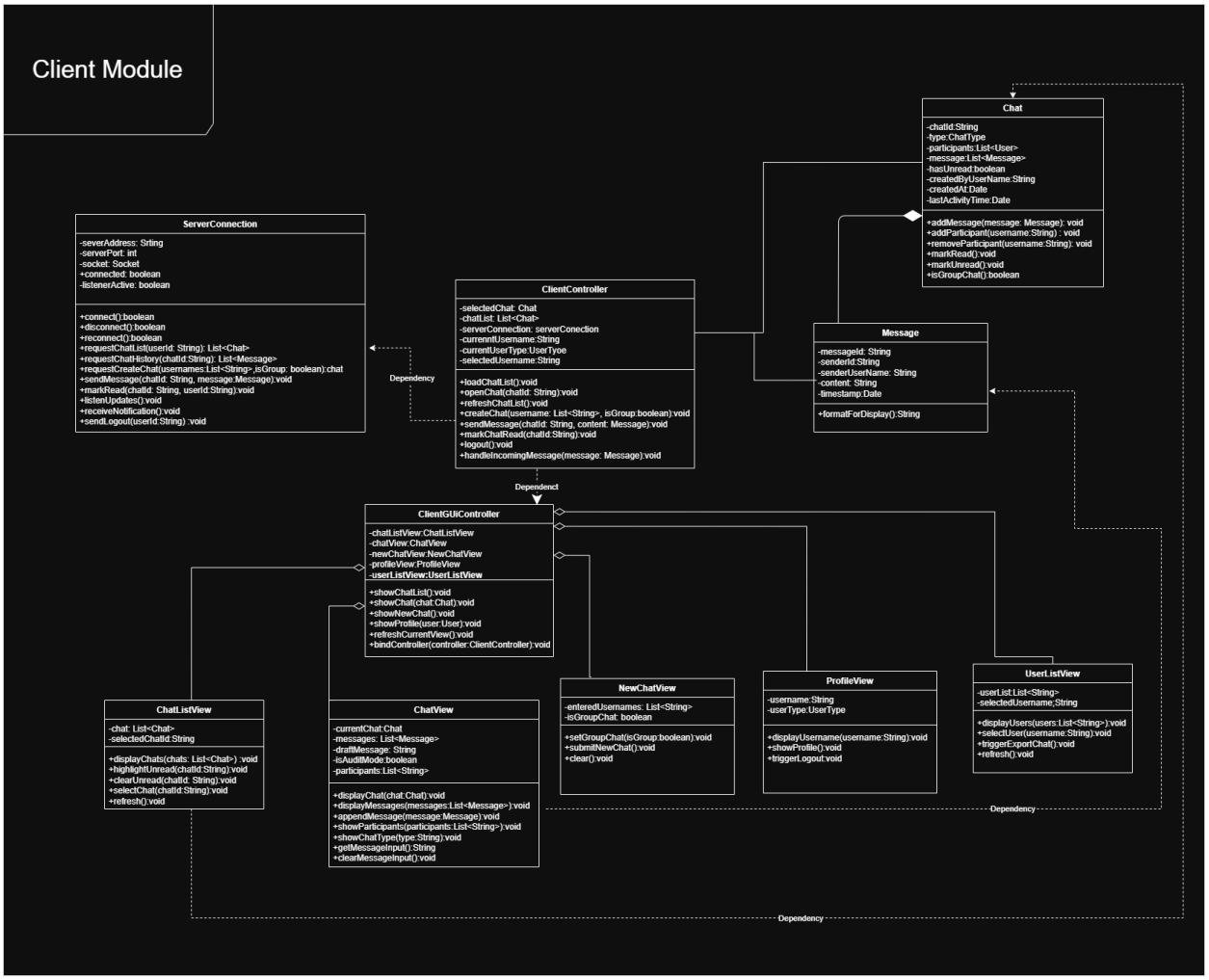
2.2. Module Architecture

2.2.1. User Login Module



The ULM provides the interface of login and handles the initial authentication. It consists of a Login View for username and password input, an authentication request handler for sending credentials to the CSM, and a Role Controller for routing authenticated users to CM or IM. The ULM does not perform authentication, it just acts as the entry point for credentials submission and response handling.

2.2.2. Client Module

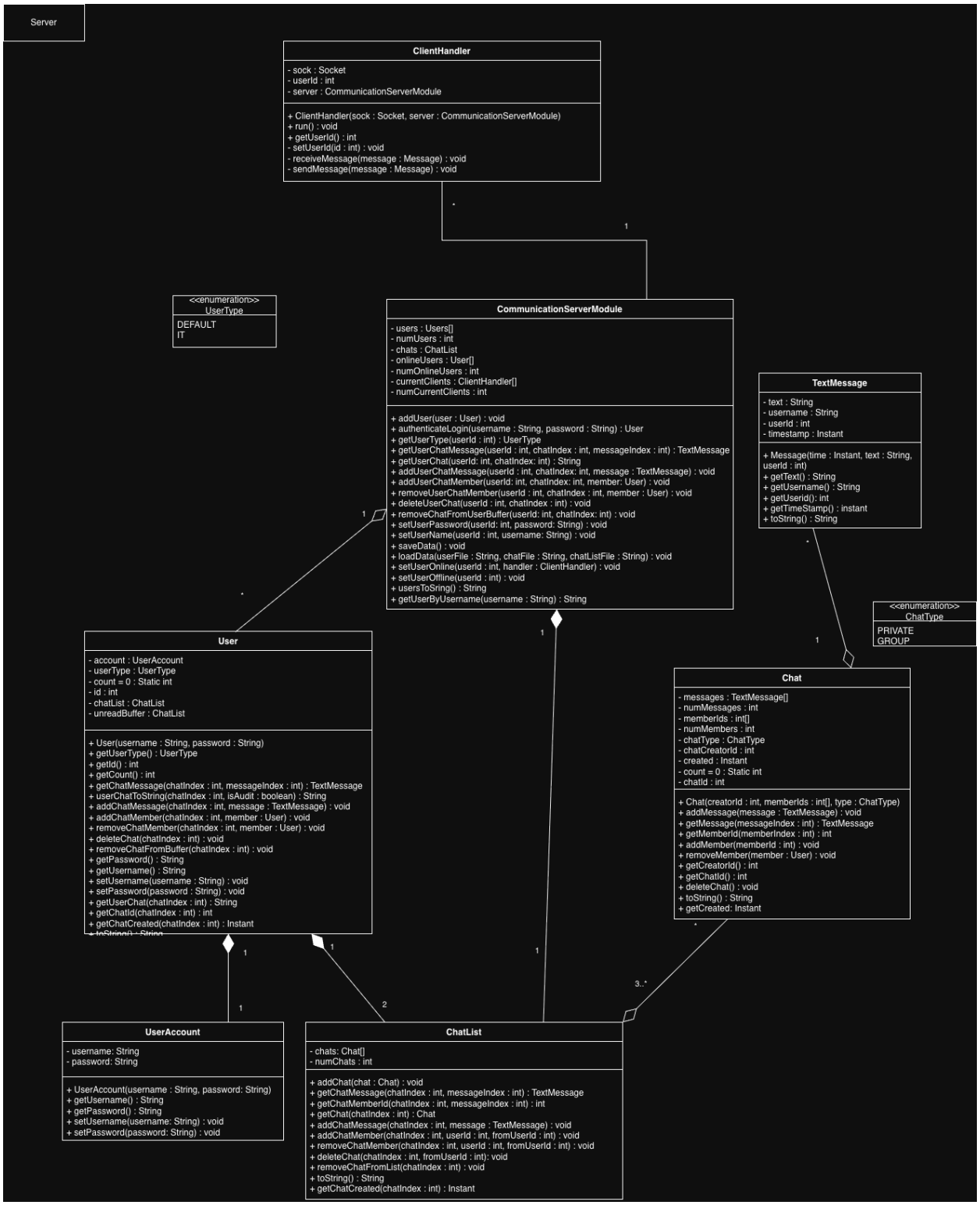


The CM module provides the user communication interface. It consists of a Chat List component, Chat View component, Message Composer component, Profile component, View UI Controller component, and a Client Network Handler component. The Chat List component displays the user's chats. The Chat View displays the full message history of selected conversations. The Message Composer catches user texts and sends message requests to the CSM. The Client Network handler maintains the TCP/IP connections and processes server updates like new messages and refreshed chat lists.

2.2.2.1. IT Sub-Module

For IT users, the capabilities of CM are extended by adding audit mode. It includes all communication features available to CM, plus an Audit Menu User search/List component, Audit Chat Viewer, and Chat Export component. The IM allows IT users to inspect conversation in read-only mode and export chat log locally. write operations such as sending, deleting, or editing message are disabled while in Audit mode,

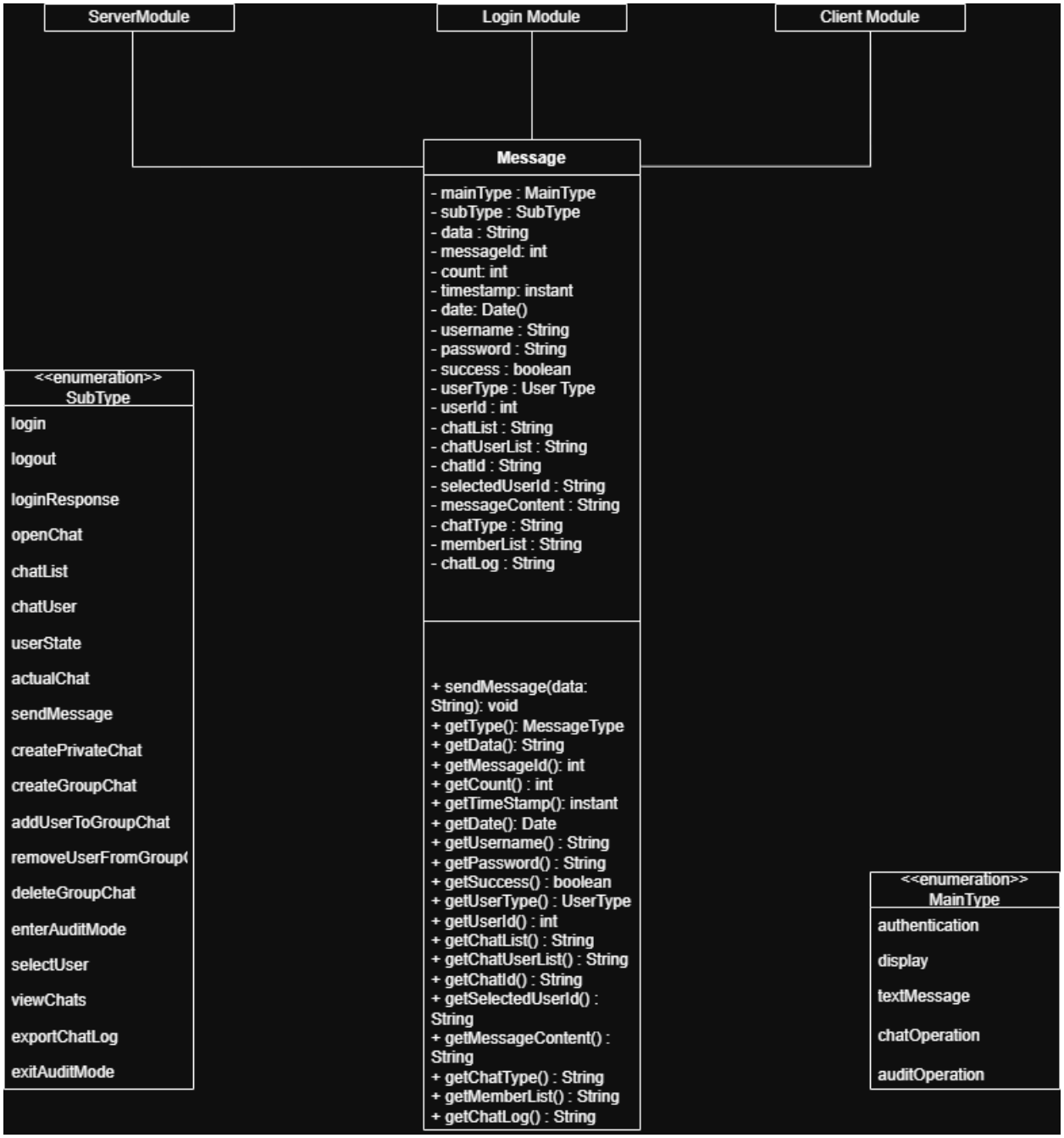
2.2.3. Communication Server Module



The Communication Server Module is the logical center of the System. It takes incoming messages through the client's specific ClientHandler to interact with the CSM through its public interface. The CSM is an aggregation of users, contains a chat list to keep track of all chats, and

has a handle on each active user's ClientHandlers to allow interaction with user clients. A chat is an aggregation of each TextMessage sent by a user to a specific chat and ChatList is an aggregation of chats. The User class consists of an account which encapsulates login information, and 2 chatlists to keep track of the chats as well as which chats are unread.

2.2.4. Message Class



The Message Class takes care of all the main and sub types of messages sent on the display. The main type of messages include authentication, display, textMessage, chatOperation, and auditOperation. The authentication message includes the login/logout part. It asks the user to

enter a username and a password, and then displays whether or not the login is successful. The display shows chat list, chat user list, and message content. The text messages are the specific messages sent between users. The chat operation messages include creating private or group chats, adding or removing users from group chats, and deleting chats. It will carry information like chat id, user id and member list. The audit operation messages represent the administrative actions. These messages are used to monitor and review system activities.

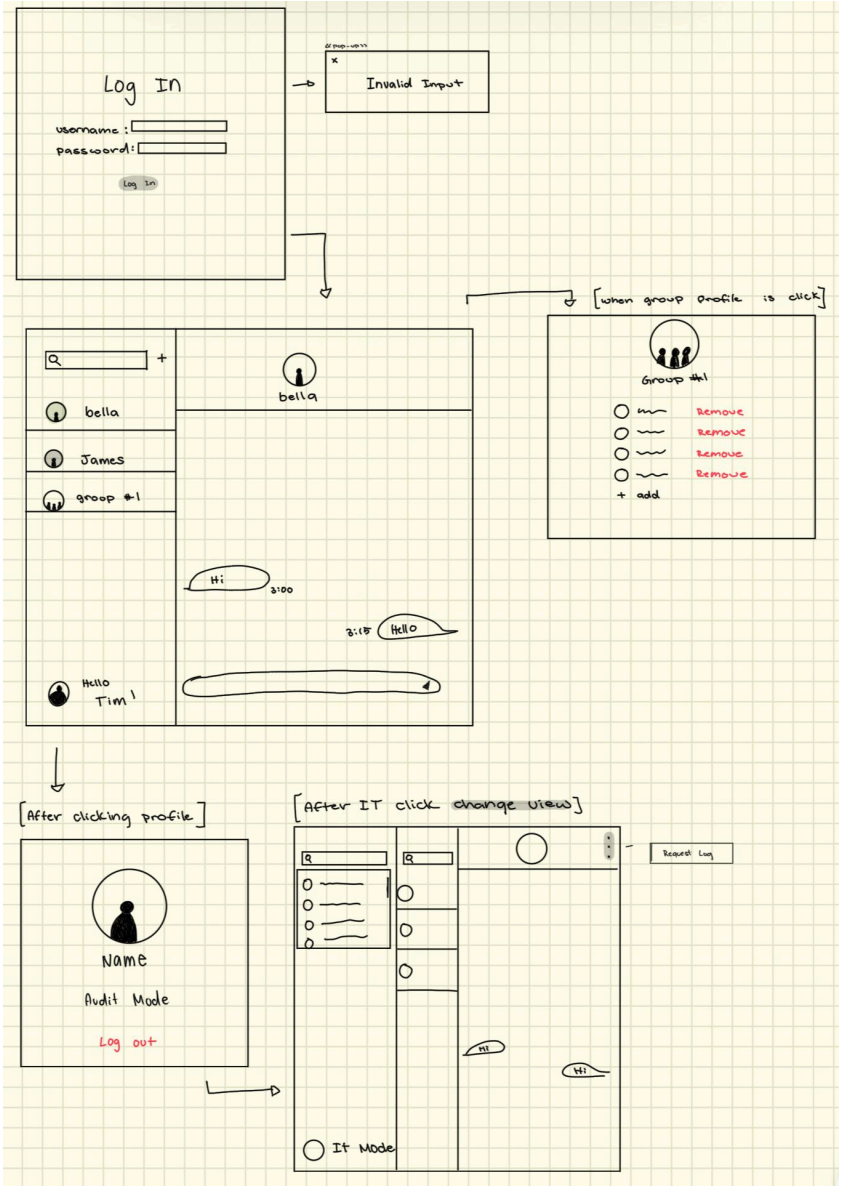
3. Data Design

3.1. Data Description

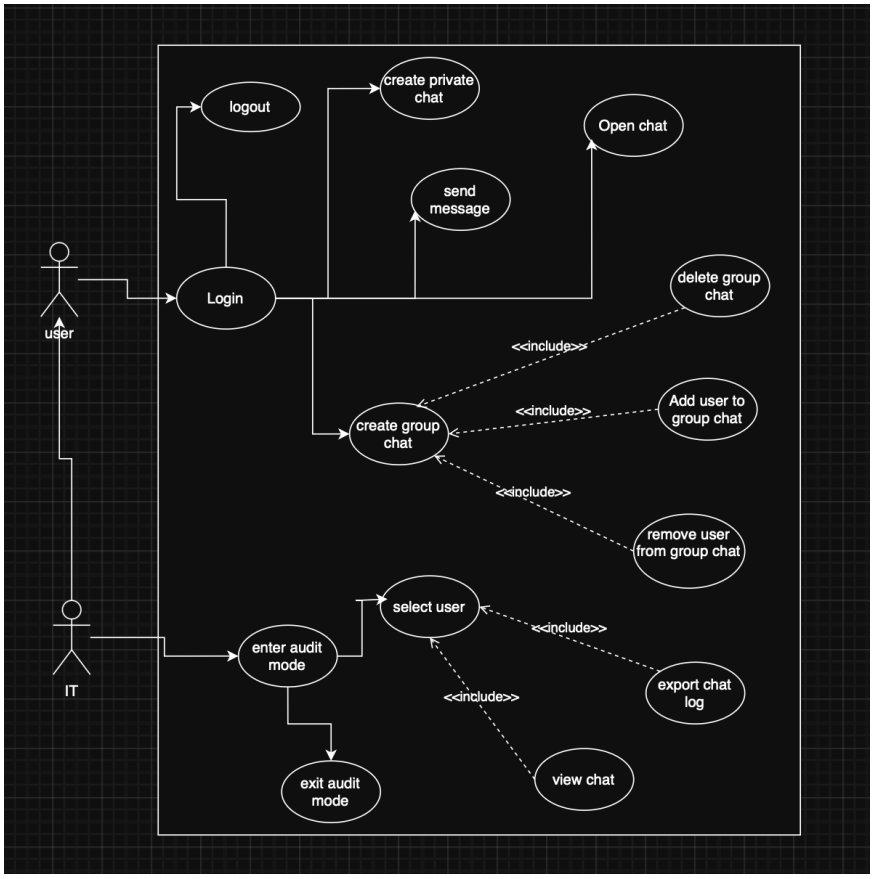
The data will be stored locally to the server in text files. A user list text file storing user data in user id, user type, username, and password separated by commas. Each line in this file is a different user. The chat list text file contains a list of all chats, and their ids. Each chat has its own file named “id”.txt, and the first 4 lines will contain the member ids (separated by commas), chat type, created date, and creator id separated by newlines. All lines after that will be the message history. This is done so the server can save and load this information for different sessions of the program running.

4. User Interface

4.1. GUI + Use Case overview



The figure above shows the GUI design for our communication app. When users open the app, they are presented with a login screen where they enter their username and password. After a successful login, they are taken to the main chat screen, where they can view all conversations and access the app's core chat features. This screen includes several clickable buttons that allow users to perform actions such as opening existing chats and starting new conversations. IT users can switch modes by clicking on their profile and selecting Audit View. In this mode, they can select a user, review that user's chats, and export chat data

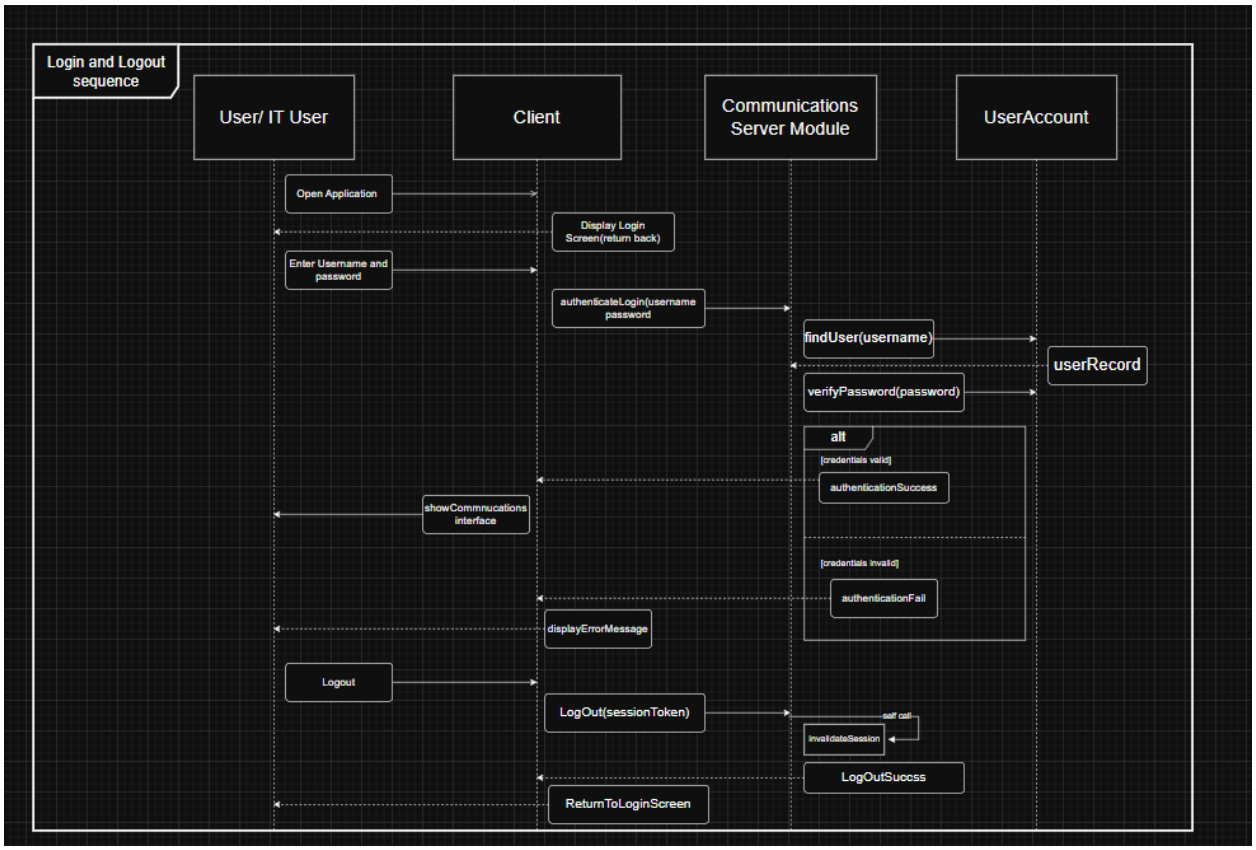


4.2. Use Cases

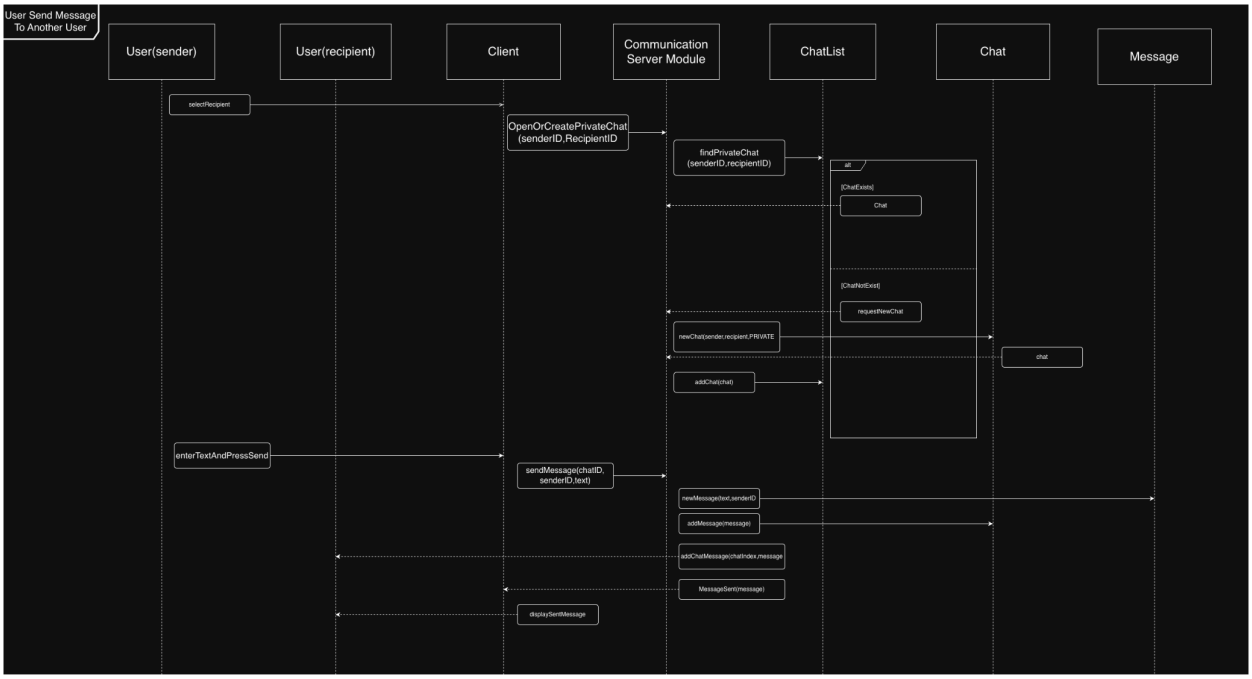
See Use Case Specification

5. System Interaction

5.1. Login/Logout Sequence

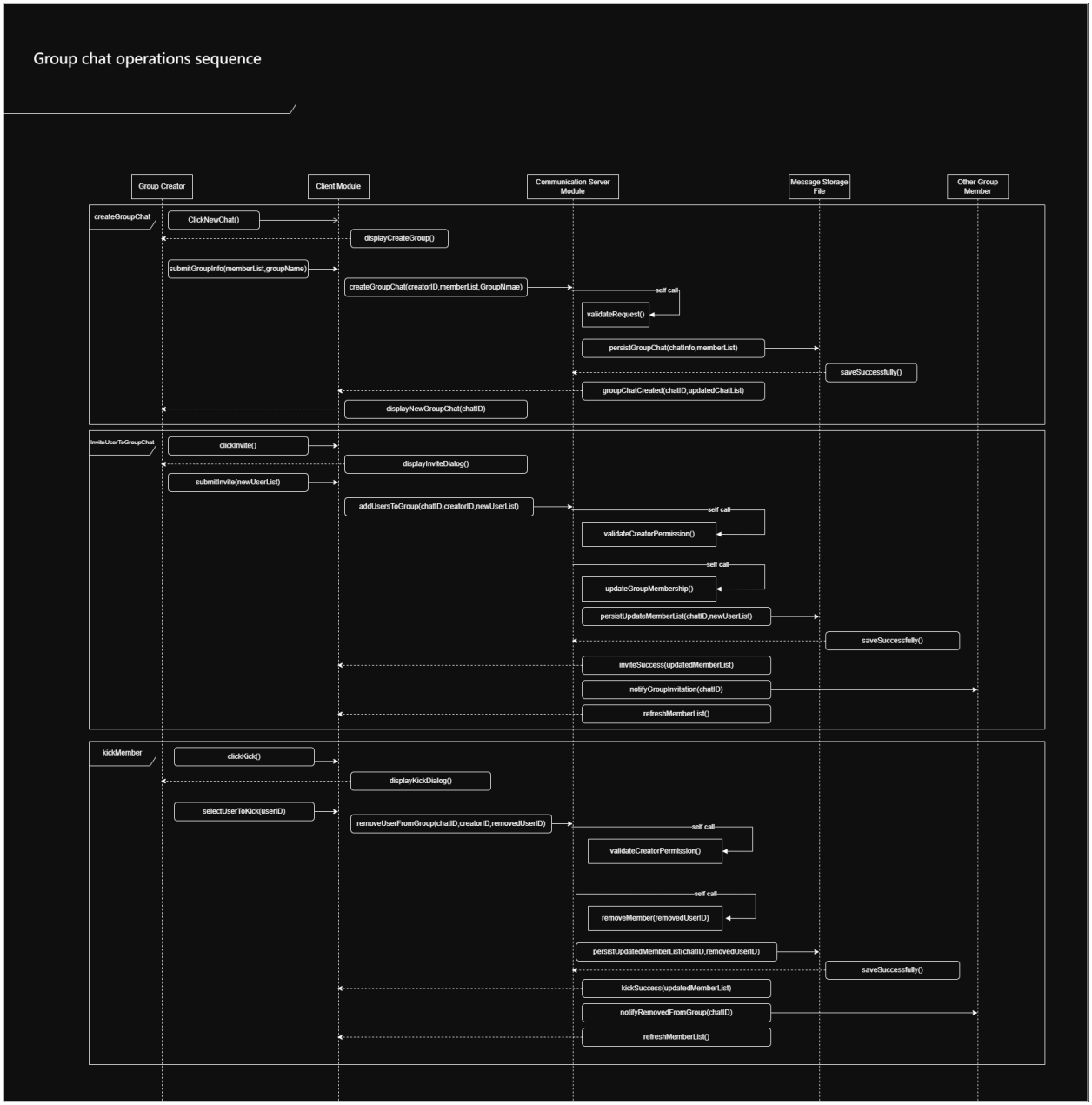
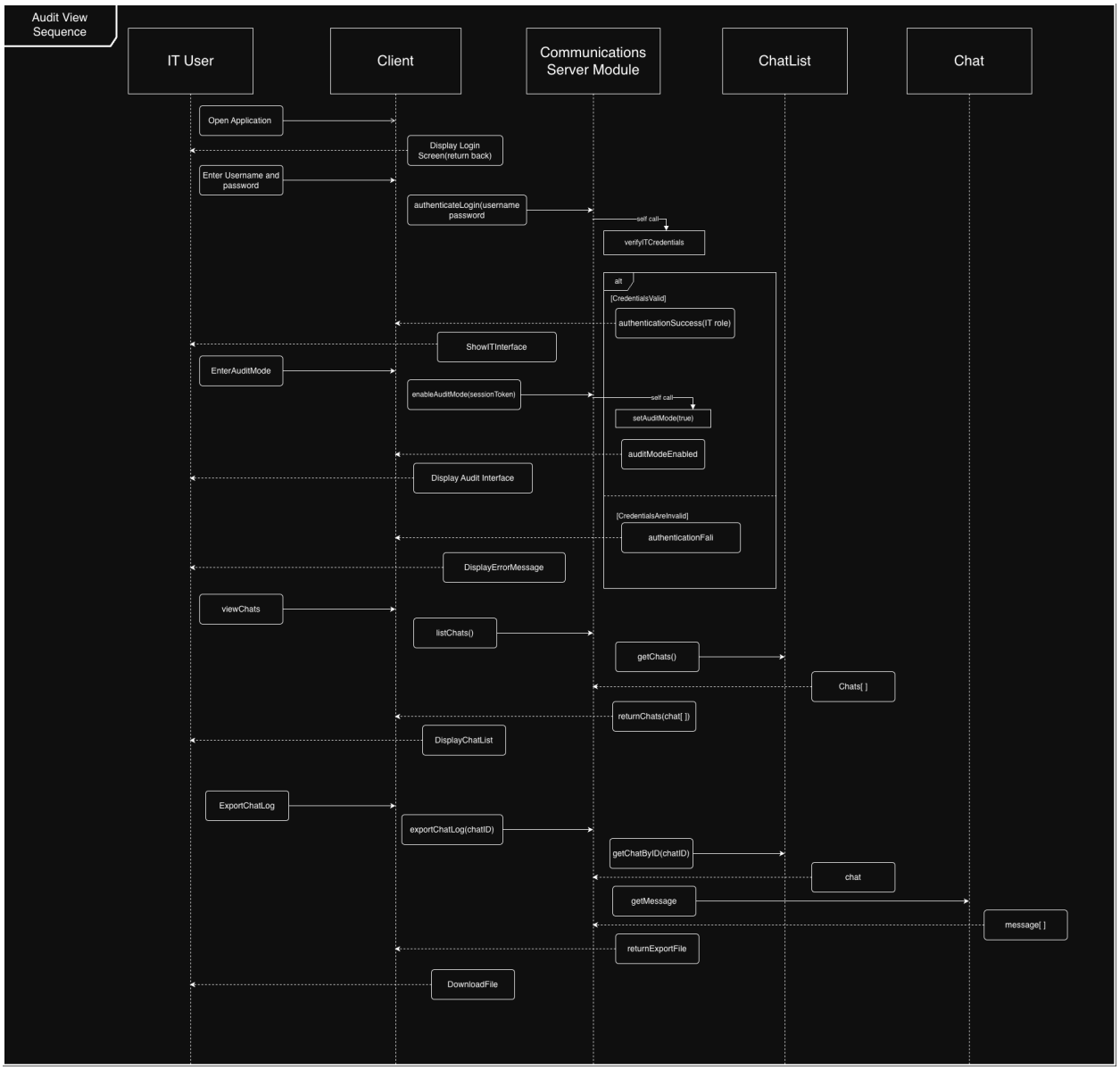


5.2. Synchronous Message Sending Sequence



5.3. Asynchronous Message Sending Sequence

5.4. Group Chat Operations Sequence



5.5. Audit View Sequence